

Boston House-Pricing Prediction using Simple and Advanced Regression Analysis

Charles Fredric G. Inventado
College of Information
and Computing Sciences
University of Santo Tomas
Bulacan, Philippines
charlesfredric.inventado.cics@ust.edu.ph

Abstract— Understanding the influential factors to house pricing is indeed critical for real estate businesses strategies and tactics. The study tackles about finding the features from various socio-economic and environmental factors that could affect the median house-pricing on state of Boston. This approach for this study is to conduct an Advanced Regression (Ridge, Lasso, and Elastic Net) analysis, with simple linear regression as the baseline to predict the house-pricing values and evaluate the performance of the models. Regularization and Hyperparameter Tuning aided to achieve the best results for the parametric values MSE, RMSE, and R^2 by minimizing the errors while maximizing the R^2 . The only factor that strongly increases the median-house price is currently the number of rooms per dwelling in a room (RM) with coefficient values greater than positive 1 on most regression model results. Inversely, there are four features that have significant effect on lowering the median house-price which are low-status of a population (LSTAT), distance of the house from Boston employment centers (DIS), the age of the homes (AGE), and ratio of students and teachers on a school (PTRATIO). With correlations ranging from negative 0.6 to negative 1.20.

Keywords — Data Preprocessing, Prediction, Regression Analysis, Simple Linear Regression, Advanced Models Regression, Hyperparameter Tuning, Overfitting, Regularization

I. INTRODUCTION

On this study, the researcher will predict the median-value of owner-occupied homes in Boston using the Boston house-price dataset which originated from the study conducted by Harrison, D. and Rubinfeld, D.L.

A. Problem

The core problem for this study is to find methods on using multiple variables for regression analysis and find optimal results of regression models that could predict median house-pricing of houses with good performance with objective metrics that would be discussed on methodology.

The first problem to be solved from this study is to determine which records from the Boston Dataset does contains outlier values that might hinder on getting a better predictive value for Median value of owner-occupied homes (MEDV).

This will also follow to determine which columns can be removed, considered as the non-relevant features that might not be needed or cause problems on the evaluations of the model. Examples of the issues that can be faced are multicollinearity and weak correlations to the target variable.

The study will also need to formulate regression models that could predict the target value and observe how the different metrics such as MSE, RMSE, and R^2 behaves and change from Simple Linear Regression to Advanced

Regression Models such as Ridge, Lasso, and Elastic Net Regression. The coefficients of the features will also be observed from the advanced regression models.

The researcher needs also to find the most optimal values that could lead to the result on model evaluation on the advanced regression models such as alpha and l-ratio through hyperparameter tuning.

Lastly, the researcher needs to explain and articulate a meaningful interpretation of the model's regression results and to draw accurate conclusions and valuable insights regarding the feature importance and model performance.

The researcher needs also to find which features are influential and helpful on predicting the house-price value on Boston.

B. Dataset

The dataset consists of 14 columns, wherein 13 of these are the independent variables or also called features, these includes :

Column Name	Definition of the Input Variable
CRIM	Per capita crime rate by town.
ZN	Proportion of residential land zoned for lots over 25,000 sq.ft.
INDUS	Proportion of non-retail business acres per town.
CHAS	Charles River dummy variable (1 if tract bounds river; 0 otherwise).
NOX	Nitric oxides concentration (parts per 10 million).
RM	Average number of rooms per dwelling.
AGE	Proportion of owner-occupied units built prior to 1940.
DIS	Weighted distances to five Boston employment centers.
RAD	Index of accessibility to radial highways.
TAX	Full-value property-tax rate per \$10,000.
PTRATIO	Pupil-teacher ratio by town.
B	Calculated as $1000(Bk - 0.63)^2 / 1000(Bk - 0.63)^2$, where Bk is the proportion of Black residents by town.
LSTAT	Percentage of lower-status population.

Table 1. List of Input or Independent Variables.

Then there is one dependent or called target variable which includes :

Column Name	Definition of the Output Variable
MEDV	Median value of owner-occupied homes in \$1000's.

Table. 2. List of Target or Dependent Variables.

The dataset currently contains 13 numerical data type and one categorical data type which is named CHAS, wherein CRIM, ZN, INDUS, NOX, RM, AGE, DIS, TAX, PTRATIO, B, LSTAT, and MEDV columns are of float64 while the CHAS and RAD columns are of int64.

CRIM	float64
ZN	float64
INDUS	float64
CHAS	int64
NOX	float64
RM	float64
AGE	float64
DIS	float64
RAD	int64
TAX	float64
PTRATIO	float64
B	float64
LSTAT	float64
MEDV	float64
dtype: object	

Fig. 1. List of data types from the raw data of both independent and dependent variables

C. Objectives

The procedure serves as the objectives to achieve the goal of this study. It starts with exploration analysis of the data. This is defined as investigating the dataset and determining the crucial variables that could improve the performance of the model. [1]

The second step is the pre-processing of data. Which transforms the raw data into a usable format that can be used by a model for regression. This prepares data for training of machine learning models. Making sure the input data is of high quality and suitable for the chosen Machine Learning Algorithm. [2]

Third, is the implementation of the model which will vary from the simple linear regression model to advanced models (implement ridge, lasso, and elastic net regression), up to hyperparameter tuning.

Fourth, the results on the performance of the different regression models then be evaluated using the following metrics such as Mean Squared Error (MSE), R-Squared. (R^2) The criterion of these metrics will be explained in detail in the methodology section.

Fifth, the interpretation of the linear regression models on how the selected features impact the target variable. This also includes the discussion of importance on making regularization in reducing overfitting and insights that will help other future researchers who will analyze this study.

Lastly, to have well-documented documents detailing the methodology, experiments, results, and conclusions. Producing a comprehensive report of the study using the IEEE format.

II. METHODOLOGY

In this section includes the detailed procedures tailored to solving the problem of the study before, during, and after the linear regression model is being implemented.

A. Data Preprocessing

Exploratory Data Analysis

Through the process of Exploratory Data Analysis (EDA) researcher first visualize all of the variables including the target variable MEDV. The used graph is the pair plot, the image is huge and may not be clearly visible here on the document, it is viewable on the code python file provided.

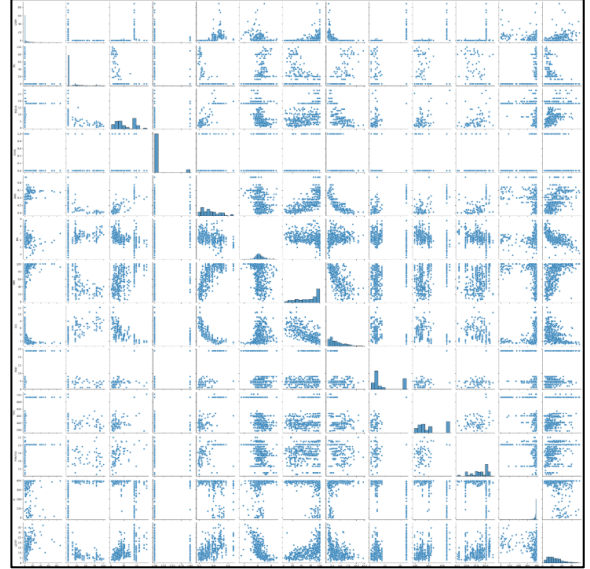


Fig. 2. Pairplot of all the Variables from Raw Data

From the visual judgement, features NOX and AGE, NOX and DIS, DIS and AGE, LSTAT and NOX, LSTAT and RM, LSTAT and DIS, INDUS and LSTAT observe to have a strong linear correlation.

This will further be investigated objectively using:

Correlation Matrix:

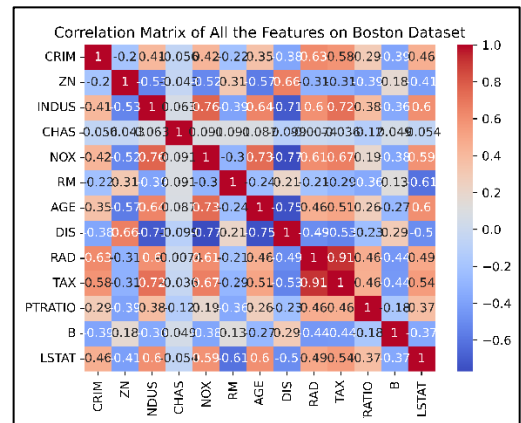


Fig. 3. Correlation Matrix of each variables from the raw data including of independent variables

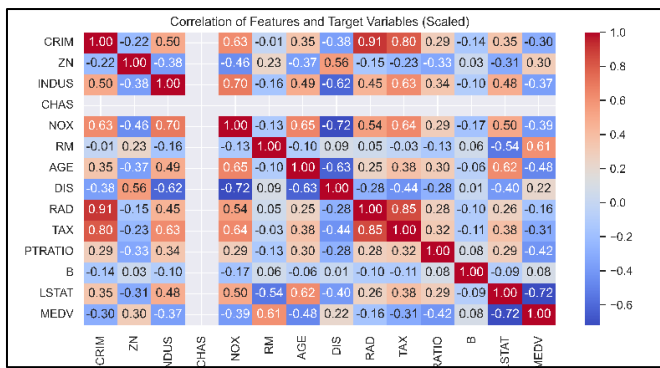


Fig. 4. Transformed and Scaled Correlation Matrix of each variables from the raw data including of both independent and dependent variables

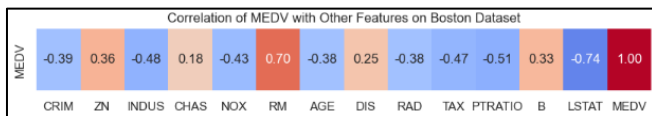


Fig. 5. Correlation Matrix of dependent variable MEDV to each other features from the raw data.

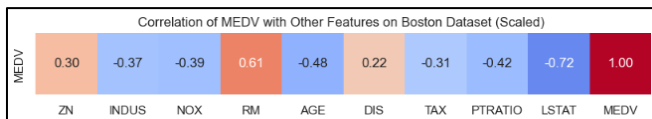


Fig. 6. Correlation Matrix of dependent variable MEDV to each other features from the transformed and scaled data.

Correlation matrix is important on deciding the justified set of features on feature selection. It also matters for both between two features and features to target variable because it determines the relationship between the two selected variables.

Continuation of Correlation Matrix Discussion:

It is crucial to determine the correlation coefficient on which the study determines the strongly correlated variables. If it is between two features, with at least ± 0.80 value [3] It is essential to deal with the two variables and on the process must at least one of the two variable removed on this case. [4]

	RAD	TAX
RAD	NaN	0.910228
TAX	0.910228	NaN

Fig. 7. Correlations on which relationship between variables is ≥ 0.80 .

From Fig. 7, features RAD and TAX have potential for multicollinearity because of its strong correlation. From Fig. 5, RAD have -0.38 correlation meanwhile the TAX have 0.47. Using the absolute value, TAX still have the higher correlation to MEDV. Furthermore, it is the decision of the researcher to remove TAX as a feature for predicting MEDV values (see feature selection)

Next part is identifying the irrelevant features from the dataset. The researcher understands the problem of having issues on multicollinearity on the experimentations by having too many features. This is done with finding correlations of MEDV to other features with threshold of ± 0.35 and below. From Fig. 5, CHAS with 0.18, and B with 0.33. Thus, with

weak correlation, the researcher has decided to remove features CHAS and B for experimental use. The researcher decided to pursue with DIS as to not remove a lot of features.

Last correlation analysis on dropping columns, Fig. 4, when the chosen columns were latter transformed and scaled, there is an obvious high correlation from CRIM to both RAD and TAX, with values 0.91 and 0.80, respectively. Furthermore CRIM can strongly influence the features. Upon decision, CRIM will also be removed for experimentation. (see feature selection)

This is the summary of the Features whether they it be used or not on experiments to predict MEDV:

Included: ZN, INDUS, NOX, RM, AGE, DIS, TAX, PTRATIO, LSTAT

Excluded: B, CHAS, RAD, CRIM

Handling Missing Values

With the appropriate command, the researcher found out that the original dataset does not contain missing value. Such that for example a 0 on CHAS does still indicate a meaning for the mentioned variable. The following figure shows the number of missing values :

CRIM	0
ZN	0
INDUS	0
CHAS	0
NOX	0
RM	0
AGE	0
DIS	0
RAD	0
TAX	0
PTRATIO	0
B	0
LSTAT	0
dtype:	int64

Fig. 8. List of missing values of all features and the target value.

Handling Outliers

Initially the researcher pondered whether he should remove the records that contain outliers from features or maybe on the target variable. He chooses to plot a Boxplot to visualize the outliers on different features and target variables.

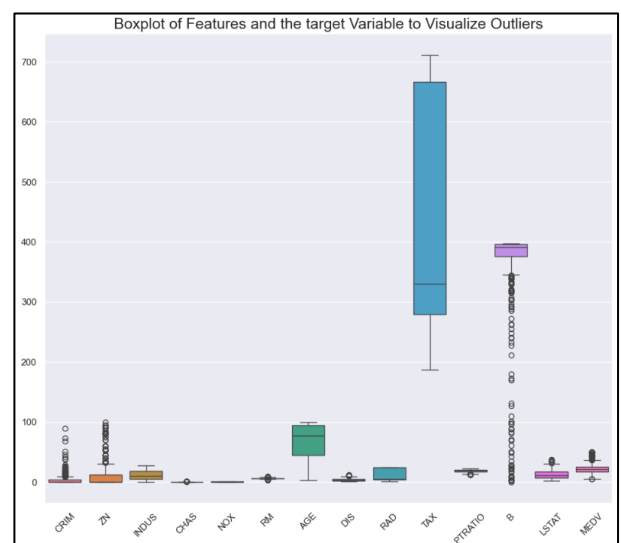


Fig. 9. Boxplot of Outliers of Feature Variables and Target Variables

From Fig. 9, CRIM, ZN, CHAS, RM, DIS, PTRATIO, B, LSTAT, and MEDV have “tiny circle” that represents outliers.

```

Outlier counts per feature:
CRIM      66
ZN        68
INDUS     0
CHAS      35
NOX        0
RM        30
AGE        0
DIS        5
RAD        0
TAX        0
PTRATIO    15
B          77
LSTAT      7
MEDV      40
dtype: int64

Features with many outliers: ['CRIM', 'ZN', 'B']

```

Fig. 10. Experiment Removing Outliers from Independent Variables

The researcher experimented with the count of the data. The first case is with the top 3 most-dense number of circles from Fig. 10, which are CRIM, ZN, and B. If we remove the records that contain outliers from any of these three features.

The researcher is bothered by the huge amount of reduction on records if he chooses between the top three independent variables. So, he chooses another case where it only includes the outliers on the target variable MEDV column.

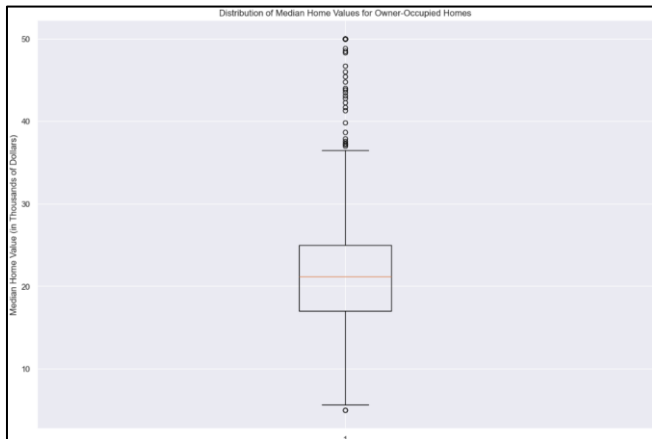


Fig. 11. Boxplot distribution of MEDV to Determine Outliers

From Fig. 10, and analysis of Fig. 11 there are numerous amount of outliers on the upper quartile of the boxplot. From Fig. 11, there are 40 outliers and the researcher decided to pursue with this decision.

Thus, the study will proceed with 466 records on the experimental part.

Scaling and Encoding

Machine learning models like regression need normalized data to provide accurate results. For continuous data such as CRIM, ZN, INDUS, NOX, RM, AGE, DIS, RAD, TAX, PTRATIO, B, LSTAT is standardized using StandardScaler() Meanwhile for categorical data such as CHAS, LabelEncoder() has to be used to standardize the values of each category on which CHAS only have two options which is binary with 1 means true and 0 being false [5].

B. Feature Selection

For feature selection, the correlation analysis technique is used. Where higher correlation to the dependent variable is deemed of higher importance to be used on the experiment. Meanwhile the independent variables that have high correlation from another independent variable can lead to redundancies and face issues such as overfitting and poor model performance [4]

For a more detailed explanation on removal of these columns, proceed back to the *continuation of correlation Matrix discussion* section.

The summary of the dropped features:

Feature Name	Basis	Description
B	Fig. 3	Weak correlation, 0.33
CHAS	Fig. 3	Weak correlation, 0.18
RAD	Fig. 4	High correlation with TAX, 0.85
CRIM	Fig. 4	High correlation with TAX and RAD, 0.91 and 0.80, respectively

Table. 3. List of Dropped Features.

These columns CRIM.CHAS, RAD, and B will be dropped with an appropriate command.

After the remaining features are also transformed and scaled, the data is now available to be fed on regression models.

C. Model Implementation

The study used different regression models to conduct the experiment analysis.

Baseline Model :

- 1.) **Simple or Multiple Linear Regression** – This model will serve as a basis for the changes to occur on the data on advanced regression models.

Advanced Regression Models :

- 2.) **Ridge Regression Model** – L2 Regularization, advanced linear regression model to address overfitting. Contains alpha values to control the regularization of the data. The values of coefficient will not be able to shrink to zero.
- 3.) **Lasso Regression Model** – L1 Regularization, reduces errors caused by overfitting like on Ridge. Alpha value minimizes the MSE or other performance metrics.

Increase on model bias and decrease on variance on higher values of alpha. Also, coefficients can shrink to zero on large values of alpha.

- 4.) **Elastic Net Regression Model** – Combination of L1 is Lasso , L2 is Ridge representing their penalties. 0.5 l-ratio gives a balance of both regression types.

Optimize parameter values :

5.) Hyperparameter tuning – GridSearchCV -

Grid search does the combination of hyperparameters into the model and gives us the best set of values. Conducting the evaluation with hyperparameter tuning is made easier and we are more confident we get good values of parameters than manual inputs of parameters.

III. EXPERIMENTS

This section includes the objective metrics and important steps to be determined within the experiment properly.

Performance Metrics Measures

These are the specific measures to be monitored that is crucial on interpreting the evaluation of the regression model result.

- 1.) **Mean Squared Error (MSE)** – The average squared difference of the predicted values to the actual values.

The MSE ranges from 0 up to any positive value where 0 indicates that the model has no error. [6]

- 2.) **Root Mean Squared Error (RMSE)** - Average difference between a model's predicted values and actual values. It is known also as the standard deviation of the residuals, which refers to the difference between the regression line and data points. RMSE can range from zero to positive infinity [7]

- 3.) **R²** – Proportion of variance for a dependent variable that is explained by an independent variable. [8]

The value of R² ranges from 0 to 1 where 1 indicates a perfect fit to the actual data.

Splitting of Training and Testing Data

The standard splitting percentage will be used, 80/20, for the training and testing of data, respectively throughout the experiment.

Hyperparameter Tuning

GridSearchCV usage on Ridge and Lasso Regression models with regularization parameter α increments from 0.01 to 0.001 and records the MSE, RMSE, and R². The best α value parameter is selected based from the value of minimizing for the MSE and RMSE while maximizing for the R²

IV. RESULTS AND ANALYSIS

This first part of this section will be divided into three parts for clarity ; second part includes discussing importance of regularization, discussing the coefficients after it is fed to the models and which features most influenced on predicting house prices :

First batch naming it A amongst comparison of models regards on the dataset with filtered outliers and filtered columns, without hyper tuning parameters conducted.

Second batch naming it B amongst comparison of models regards on the dataset with filtered outliers and filtered columns, with hyper tuning parameters conducted.

Lastly, batch C amongst comparison of models regards on the dataset without filtered outliers and columns. Without hyperparameters as well. Thus, the batch with raw data.

A. BATCH A (Without Hyperparameter Tuning)

Regression Model	MSE	RMSE	R ²
Simple Linear Regression	5.07	2.25	0.80
Ridge Regression	5.28	2.30	0.79
Lasso Regression	5.71	2.39	0.77
Elastic Net Regression	5.66	2.38	0.77

Table. 4. Results of Batch A (With Filtered Outliers, without Hyperparameter Tuning)

Although the simple linear regression does not have advanced parameters, at this case on Batch A, have the best result for all parameters compared to three other advanced regression models.

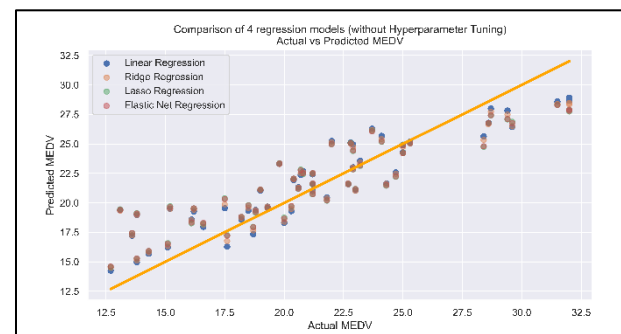


Fig. 12. Comparison of Four Regression Model Results of Batch A (With Filtered Outliers, without Hyperparameter Tuning)

The differences among the gaps of the data points of the predicted value MV from the training data does have a little difference in terms of y-value visual gaps. Having an MSE around 5 is already considered to be small, the model is predicting precisely.

Next, we will proceed to having hyperparameter tuning on the regression models, with a goal to further minimize the MSE and RMSE, then maximizing as well the R²

B. BATCH B (With Hyperparameter Tuning)

Regression Model	MSE	RMSE	R ²
Simple Linear Regression	5.07	2.25	0.80
Ridge Regression	5.09	2.26	0.80
Lasso Regression	5.13	2.27	0.80
Elastic Net	5.68	2.38	0.77

Regression			
------------	--	--	--

Table. 5. Results of Batch B (With Filtered Outliers, with Hyperparameter Tuning)

The results are surprising across the advanced regression model results. The MSE, RMSE, and R^2 specifically for Ridge and Lasso Regression is close already to the value of simple linear regression. (which have not changed)

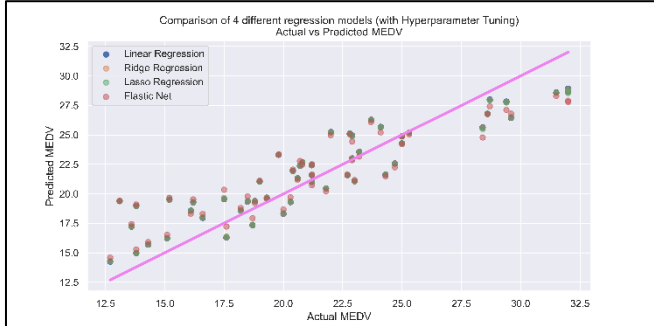


Fig. 13. Comparison of Four Regression Model Results of Batch B (With Filtered Outliers, with Hyperparameter Tuning)

Furthermore, from Table. 4 and Table. 5, we can observe that there is improvement in MSE, RMSE, and R^2 of the advanced regression models when we used the Hyperparameter Tuning.

The researcher thinks that the input of parameters, especially manual input from batch A on advanced models, is not yet the best-case scenario of alpha value and l1-ratio value for example.

C. BATCH C

(Without Hyperparameter Tuning, with Outliers)

Regression Model	MSE	RMSE	R^2
Simple Linear Regression	24.54	4.95	0.78
Ridge Regression	24.96	5.00	0.78
Lasso Regression	24.76	4.98	0.78
Elastic Net Regression	26.36	5.13	0.76

Table. 6. Results of Batch C (Without Filtered Outliers, without Hyperparameter Tuning)

There is a concerning and massive increase in the values of the parameters on MSE and RMSE for all regression models. The precision of the data points is not promising, the visualization must be prove this point.

With outliers, there lies the issue of bad performance within models. This is a basic lesson for everyone, to always remove these anomalies present on the data sets.

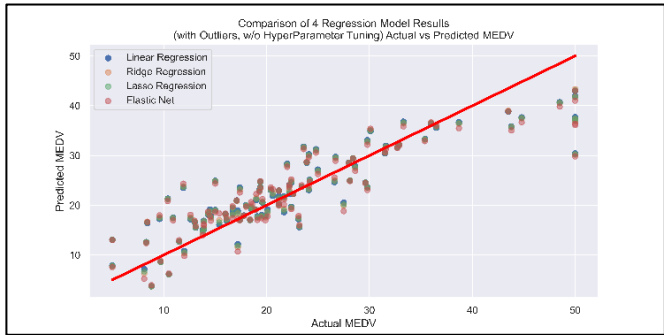


Fig. 14. Results of Batch C (Without Filtered Outliers, without Hyperparameter Tuning)

Recall that the raw data contains more records (506) compared to the reduced and transformed data set. (260) And also the raw data sets have the irrelevant columns such as CHAS and B or rather the ones with weakest correlations on the model.

Furthermore, a combination of these problems makes the predictive regression models almost useless. If the prediction is too sparse, the model and data is indeed not good.

D. Understanding Coefficients and Determining Influential Features to Median-Price Housing Value

Tackles which features are the most influential on predicting house prices and how the coefficients grow or shrink with different regression models and Hyperparameter Tuning.

Regression Model	Feature Name	Batch A (w/o HT)	Batch B (w/ HT)	Batch C (w/ outliers)
Simple Linear Regression Model	ZN	0.09	0.09	0.03
	INDUS	-0.09	-0.09	-0.06
	NOX	-0.34	-0.34	-14.09
	RM	1.48	1.48	3.75
	AGE	-1.10	-1.10	0.007
	DIS	-1.15	-1.15	-1.31
	TAX	-0.18	-0.18	-0.002
	PTRATIO	-0.86	-0.86	-0.84
Ridge Regression Model	LSTAT	-1.16	-1.16	-0.56
	ZN	0.07	0.09	0.04
	INDUS	-0.08	-0.09	-0.09
	NOX	-0.30	0.34	-7.99
	RM	1.41	1.48	3.70
	AGE	-0.99	-1.09	0.002
	DIS	-0.96	-1.12	-1.22
	TAX	-0.18	-0.18	-0.003
Lasso Regression Model	PTRATIO	-0.82	-0.85	-0.76
	LSTAT	-1.18	-1.16	-0.57
	ZN	0.00	0.07	0.04
	INDUS	-0.00	-0.07	-0.07
	NOX	-0.15	-0.33	-10.96
	RM	1.37	1.48	3.69
	AGE	-0.87	-1.08	0.005
	DIS	-0.63	-1.08	-1.26
	TAX	-0.14	-0.18	-0.002
	PTRATIO	-0.79	-0.85	-0.80
	LSTAT	-1.26	-1.17	-0.56
	ZN	0.0	0.0	0.04

Elastic Net Regression Model	INDUS	-0.0	-0.0	-0.12
	NOX	-0.21	-0.19	-0.0
	RM	1.37	1.37	3.38
	AGE	-0.88	0.87	-0.0004
	DIS	-0.68	-0.67	-1.047
	TAX	-0.15	-0.15	-0.005
	PTRATIO	-0.79	-0.79	-0.66
	LSTAT	-1.23	-1.24	-0.60

*HT = Hyperparameter Tuning

Table.7. Results of Coefficients on different Regression Models of three Different Batches

Observations from Table. 7. The Average number of rooms per dwelling (RM) has a strong positive influence on predicting house-pricing value with values consistently greater than one.

Also high on these but it lowers the median price for many of the features including LSTAT or lower-status population can tend to influence the median price-value downward of the price, DIS or distances from Boston employment centers as well can highly decrease the median of house-price, AGE of the houses or buildings also can lower the median of house-price, and PT RATIO or pupil-teacher ratio also can lower median of house-price, these have high values close to one as well ranging from 0.6 to 1.20 on absolute value.

Meanwhile, it is the opposite case these features to be mentioned do not have strong influence to the median-price value of houses. Starting with ZN or proportion of residential landed zones, INDUS or proportion of non-retail businesses, NOX or nitric oxide, and TAX, which has low coefficients ranging from 0 to 0.4 absolute value.

Also another observation on ZN and INDUS Batch A and B (where there is applied regularization) result on Ridge and Lasso Model, respectively, that it attained a value of 0.0 or it has no value at all to contribute on the house-pricing value prediction.

E. Importance of Regularization

By definition, regularization reduces the errors from the predicted value by fitting the function in a way it avoids overfitting on the training set. Regularization occurred for all of the advanced regression models. [9]

Furthermore, by adding constraints such as L1 Regularization and L2 Regularization, Lasso and Ridge Regression Respectively, and a combination of two or Elastic Net Regression will aid on accuracy of the data (such as MSE and RMSE) and have a reliably balanced training data that is not complex. Thus, regularization is helpful and can be used on day-to-day applications.

Along with Hyperparameter Tuning, combined with regularization, there can be a more optimized model performance, making advanced regression models better than simple linear regression in reality.

V. CONCLUSION

A. Learnings / Lessons

Before application of different regression models, it is important before conducting experimental proper to have proper-preprocessing of data. Like a highway, there is no shortcut to get good performance or accuracy on a model. It must proceed to different phases on where it could help improve the data from the raw dataset.

Thus, pre-processing such as data cleaning of missing values and finding outliers on the data must be the top priority of researchers.

The feature selection also is important as to view there are a lot of features initially, but variables with low correlation such as CHAS, high multicollinearity like TAX and RAD, researchers must always visualize first because it makes understanding of the data set easier so the researcher can filter out immediately the one that is deemed to seem useless for the model.

Testing out different regression models will expand one's dataset flexibility, with the aid of regularization as to avoid the complexity of the data by penalizing the coefficients which seem to not already relevant for the prediction. Then there is also Hyperparameter Tuning is similar to an automated assistant, it will give you the most optimal parameters immediately, it does not need to be complicated of finding the best value manually.

B. Key Insights

Pre-processing requires visualization process to understand which features are likely to be used or not. Such as correlation matrix analysis and other issues on variables that might cause worse performance of the model in the experiment data set. This is the cause of features such as per capita crim rate (CRIM), non-retail business population (INDUS), people living near the Charles River (CHAS), number of black residents (B) being filtered out for the experiment.

Initially, parametric values on simple linear regression gives the best results, but the experiment proceeded to attempt advanced features from Ridge, Lasso, and Elastic Net Regression such as control of the regularization strength (L1, L2) on respective models and hyperparameter tuning determining the best values for L1 and L2, these helped the researcher to conclude which features are important on predicting median house-price value in Boston.

In the end, using the filtered the number of rooms per dwelling (RM) is the only feature that could increase the median house-pricing value on the Boston dataset. There are four features that tends to lower the median-house price, the low-status of a population (LSTAT), distance from Boston employment centers (DIS), age of the structures such as houses or buildings (AGE) and ratio of students and teachers on school. (PTRATIO)

REFERENCES

- [1] Ibm. (n.d.). Exploratory Data Analysis. *What is exploratory data analysis (EDA)?* <https://www.ibm.com/think/topics/exploratory-data-analysis>
- [2] University of Arizona Global Campus. (n.d.). *What is data processing?* UAGC. <https://www.uagc.edu/blog/what-data-processing>
- [3] Frost F. (n.d.). *Interpreting Correlation Coefficients*. Statistics by Jim. <https://statisticsbyjim.com/basics/correlations/>
- [4] GeeksforGeeks. (2024, August 25). *How to perform feature selection for regression data*. GeeksforGeeks. <https://www.geeksforgeeks.org/how-to-perform-feature-selection-for-regression-data/>
- [5] <https://www.analyticsvidhya.com/blog/2020/03/one-hot-encoding-vs-label-encoding-using-scikit-learn/> solving nonlinear integro-differential equations of Fredholm type," *J. Comput. Math.*, vol. 34, no. 3, pp. 262–284, May 2016, doi: 10.4208/jcm.1512-m2015-0241.
- [6] Frost F. (n.d.). *Mean Squared Error (MSE)*. Statistics by Jim. <https://statisticsbyjim.com/regression/mean-squared-error-mse/>
- [7] Frost F. (n.d.). *Root Mean Squared Error (RMSE)*. Statistics by Jim. <https://statisticsbyjim.com/regression/root-mean-square-error-rmse/>
- [8] Fernando, J. (2024, November 13). *R-Squared: Definition, Calculation, and interpretation*. Investopedia. <https://www.investopedia.com/terms/r/r-squared.asp>
- [9] GeeksforGeeks. (2024a, August 5). *Regularization in machine learning*. GeeksforGeeks. <https://www.geeksforgeeks.org/regularization-in-machine-learning/>